Arcade games

micro:bit Course

Module: Micro:bit Game Design

MARIO KART





Lesson 3

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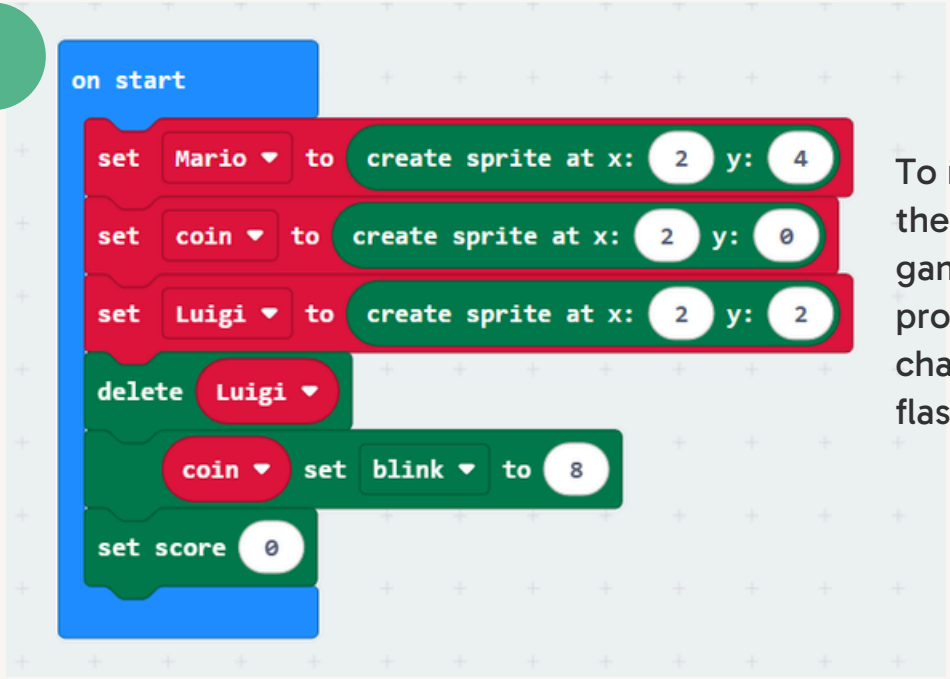


MICRO:BIT MARIO KART



In this activity you will create a Mario Kart racing game on the micro:bit! You will steer the micro:bit to move Mario out of the way and ahead of Peach, Bowser, Wario and others. See how many flashing coins you can collect!

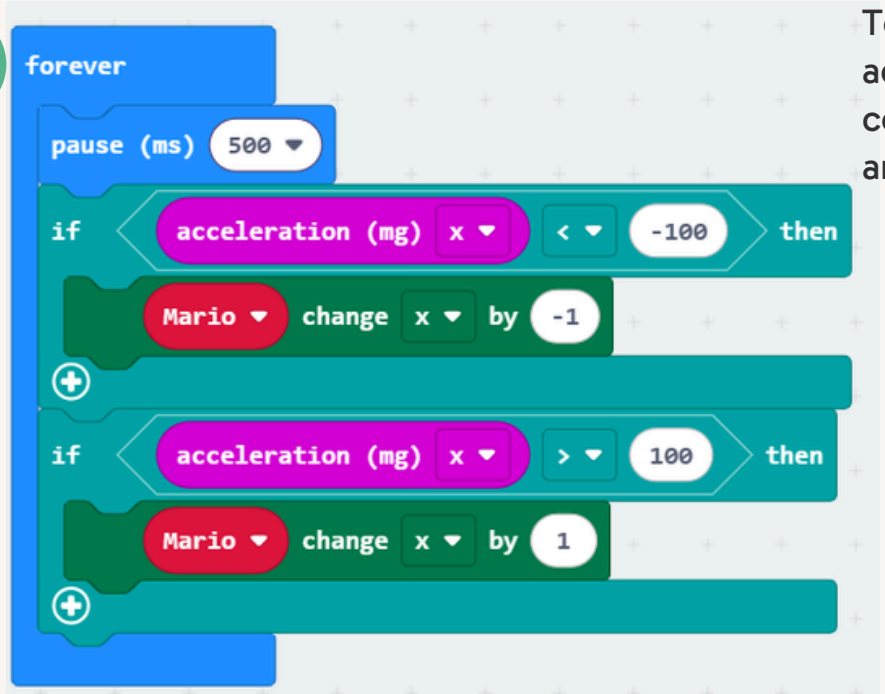
1



To make it easier to understand the code – and therefore the game – we'll start by programming just three characters: Luigi, Mario and the flashing coin.



2



To move Mario, we'll use the accelerometer again, but we could have used the two pins, A and B (which I personally prefer)

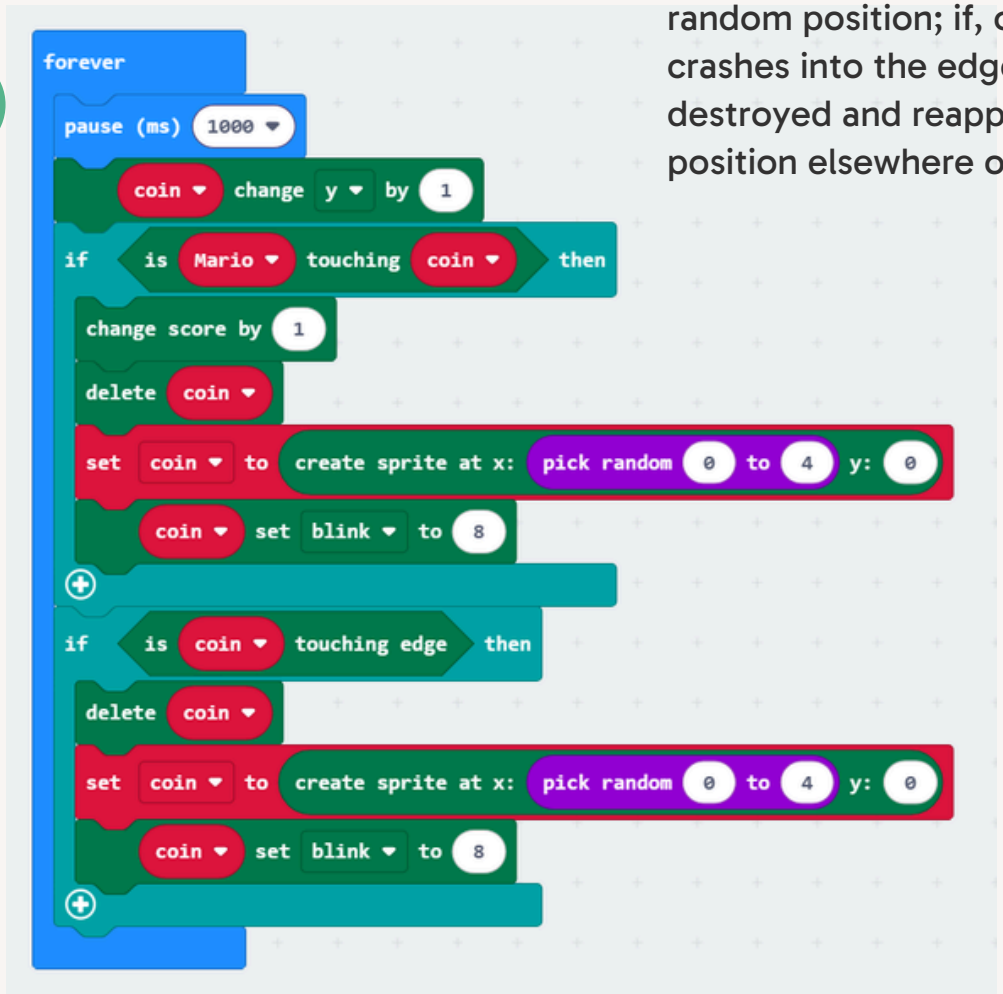


MICRO:BIT MARIO KART



At this stage of the programme, we only have two sprites: the flashing coin and Mario. All the others were destroyed at the start and then reappeared depending on the score Mario achieved.

3



TIf the coin bounces off Mario, the score increases, and the coin appears in a random position; if, on the other hand, it crashes into the edges of the screen, it is destroyed and reappears in a random position elsewhere on the screen.





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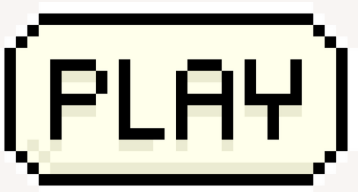
4

To move Mario, we'll use the accelerometer again, but we could have used the two pins, A and B (which I personally prefer)











MICRO:BIT MARIO KART



Now let's add all the players

1

```
on start
    set Mario to create sprite at x: 2 y: 4
    set coin to create sprite at x: 2 y: 0
    set bowser to create sprite at x: 4 y: 0
    set toad to create sprite at x: 2 y: 2
    set Peach to create sprite at x: 2 y: 2
    set Luigi to create sprite at x: 2 y: 2
    set Wario to create sprite at x: 2 y: 2
    delete bowser
    delete toad
    delete Peach
    delete Luigi
    delete Wario
    coin set blink to 8
    set score 0
```



2

```
forever
    pause (ms) 500
    if acceleration (mg) x < -100 then
        Mario change x by -1
    if acceleration (mg) x > 100 then
        Mario change x by 1
```



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3

```
forever
    pause (ms) 1000
    coin change y by 1
    if is Mario touching coin then
        change score by 1
        delete coin
        set coin to create sprite at x: pick random 0 to 4 y: 0
        coin set blink to 8
    if is coin touching edge then
        delete coin
        set coin to create sprite at x: pick random 0 to 4 y: 0
        coin set blink to 8
```



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4

```
forever
  if score ≥ 3 then
    if is Luigi deleted then
      set Luigi to create sprite at x: pick random 0 to 4 y: 0
    pause (ms) 1000
    Luigi change y by 1
    if is Luigi touching Mario then
      game over
    if is Luigi touching edge then
      delete Luigi
```



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5

```
forever
  if score ≥ 4 then
    if is Wario deleted then
      set Wario to create sprite at x: pick random 0 to 4 y: 0
    pause (ms) 1000
    Wario change y by 1
    if is Wario touching Mario then
      game over
    if is Wario touching edge then
      delete Wario
```

6

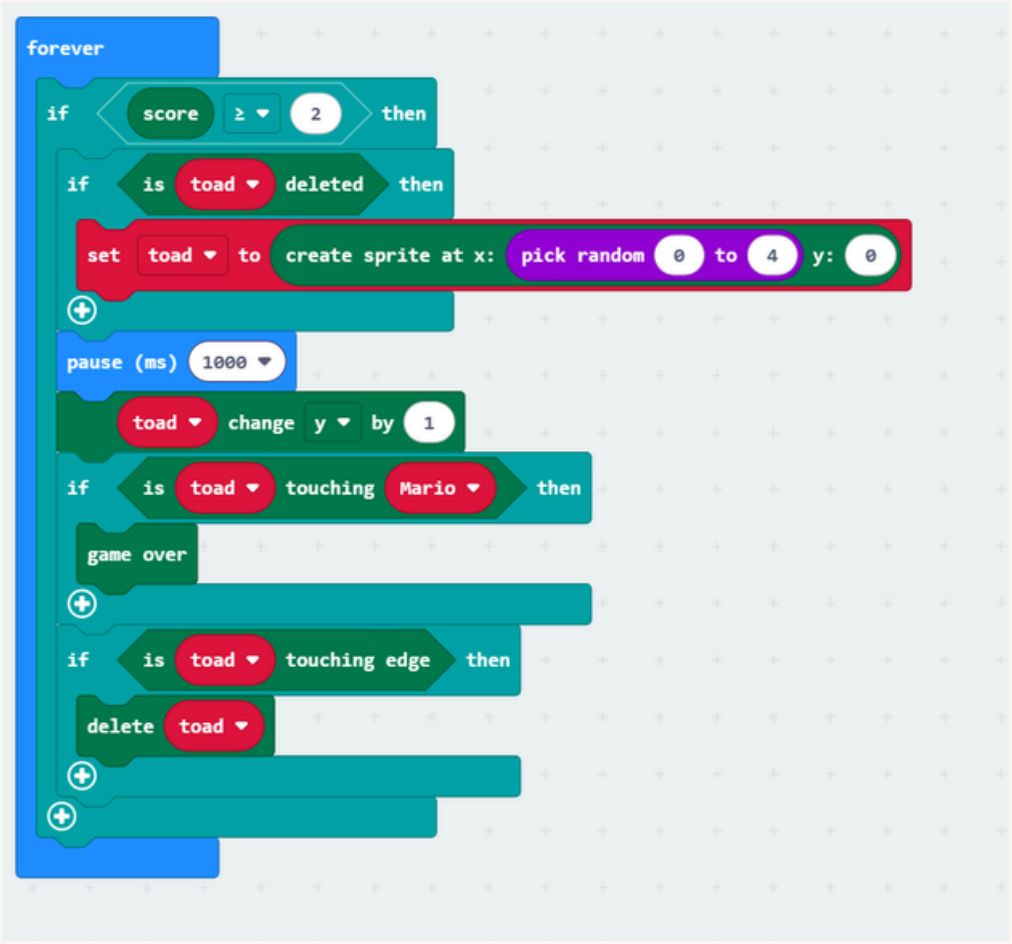
```
forever
  if score ≥ 3 then
    if is Peach deleted then
      set Peach to create sprite at x: pick random 0 to 4 y: 0
    pause (ms) 1000
    Peach change y by 1
    if is Peach touching Mario then
      game over
    if is Peach touching edge then
      delete Peach
```



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7





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8

